

**More Practice: Confusion Matrices and ROC Curves***(work with a partner)*

Throughout, “positive” is the event we want to detect. Recall:

$$\text{precision} = \frac{TP}{TP + FP},$$

$$\text{FPR} = \frac{FP}{FP + TN},$$

$$\text{recall} = \text{TPR} = \frac{TP}{TP + FN},$$

$$\text{specificity} = \frac{TN}{TN + FP}.$$

**1. Airport screening.** Of 200 bags, 20 contain a prohibited item. A scanner flags 30 bags. Of the flagged bags, 15 actually contain a prohibited item.

(a) Complete the confusion matrix, including all row and column totals.

| Predicted class | Actual class |          | Total |
|-----------------|--------------|----------|-------|
|                 | Positive     | Negative |       |
| Positive        |              |          |       |
| Negative        |              |          |       |
| Total           |              |          |       |

(b) Compute precision, recall, FPR, and specificity.

(c) In one sentence, explain what the false positives mean in this setting.

**2. Spam filter.** A test set contains 50 spam messages and 150 non-spam messages. The filter correctly sends 40 spam messages to the spam folder, but it also sends 15 non-spam messages there.

(a) Identify  $TP$ ,  $FP$ ,  $FN$ , and  $TN$ , then compute precision, recall, and FPR.

(b) Which error is a legitimate message sent to spam: a false positive or a false negative?

**Checkpoint.** Compare examples 1 and 2: which classifier has higher recall? Which has lower FPR? Show the two calculations you used.

**More Confusion Matrix Practice**

**3. Medical screening.** A screening test is evaluated on 500 people. Of the 100 people who have the condition, 90 test positive. Of the 400 people who do not have the condition, 40 test positive.

- (a) Complete the confusion matrix, including all row and column totals.

| Predicted class | Actual class |          | Total |
|-----------------|--------------|----------|-------|
|                 | Positive     | Negative |       |
| Positive        |              |          |       |
| Negative        |              |          |       |
| Total           |              |          |       |

- (b) Compute precision, recall, FPR, and specificity.

- (c) A patient receives a positive result. Which metric describes the fraction of positive results that are correct? What is its value here?

**4. Fraud alerts.** A bank reviews 1,000 transactions: 50 are fraudulent and 950 are legitimate. Its model catches 40 fraudulent transactions and incorrectly flags 20 legitimate transactions.

- (a) Complete the confusion matrix.

| Predicted class | Actual class |          | Total |
|-----------------|--------------|----------|-------|
|                 | Positive     | Negative |       |
| Positive        |              |          |       |
| Negative        |              |          |       |
| Total           |              |          |       |

- (b) Compute accuracy, precision, recall, and FPR.

- (c) The model is 97% accurate. Explain why accuracy alone can still be misleading for this problem.

**Checkpoint.** Both the medical test and fraud model have recall = 0.90 or lower. Which positive result would you trust more, and which metric supports your answer?

### Building an ROC Curve from Scores

A model assigns a risk score to eight patients. Predict positive when the score is *at least* the threshold. The rows are already sorted from highest to lowest score.

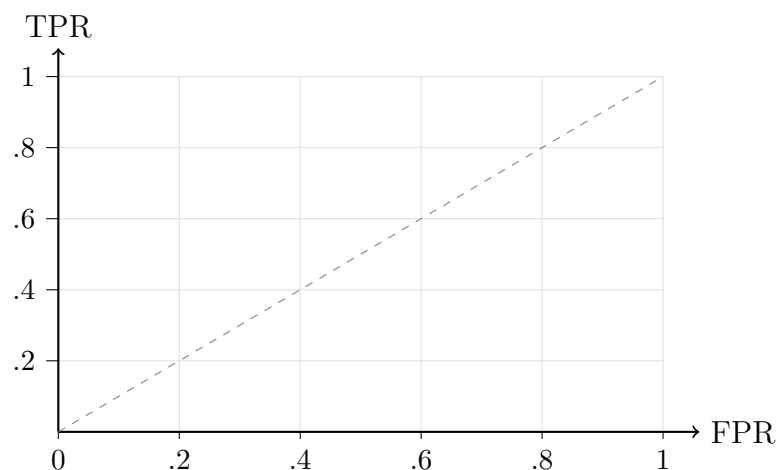
| Patient      | A   | B   | C   | D   | E   | F   | G   | H   |
|--------------|-----|-----|-----|-----|-----|-----|-----|-----|
| Score        | .95 | .85 | .75 | .65 | .55 | .45 | .35 | .25 |
| Actual class | +   | -   | +   | +   | -   | -   | +   | -   |

### 5. Thresholds change the confusion matrix.

(a) Complete the table. The first and last rows are filled in for you.

| Threshold | $TP$ | $FP$ | $FN$ | $TN$ | $TPR$ | $FPR$ |
|-----------|------|------|------|------|-------|-------|
| $> .95$   | 0    | 0    | 4    | 4    | 0     | 0     |
| .90       |      |      |      |      |       |       |
| .80       |      |      |      |      |       |       |
| .70       |      |      |      |      |       |       |
| .60       |      |      |      |      |       |       |
| .50       |      |      |      |      |       |       |
| .40       |      |      |      |      |       |       |
| .30       |      |      |      |      |       |       |
| .20       | 4    | 4    | 0    | 0    | 1     | 1     |

(b) Plot (FPR, TPR) for every row and connect consecutive points. Label at least three points with their thresholds.



(c) As the threshold decreases, what happens to TPR? What happens to FPR? Explain why.

## Introduction to Probability

For a discrete outcome  $e$ , we can estimate its probability by its relative frequency:

$$P(e) = \frac{\text{count}(e)}{\text{count}(\text{all outcomes})}.$$

**8. Sampling from a bag.** A bag contains eight tiles labeled

$$A, A, A, A, B, B, C, D.$$

One tile is selected uniformly at random.

- (a) Write the probability distribution for the label of the selected tile.
- (b) Compute  $P(\text{label is } B \text{ or } C)$ .
- (c) If the experiment is repeated 80 times with replacement, about how many times would you expect to select  $A$ ? Is that count guaranteed?

**9. Probability axioms.** A random variable  $X$  has the distribution below.

| $x$        | 0    | 1    | 2   |
|------------|------|------|-----|
| $P(X = x)$ | 0.20 | 0.50 | $q$ |

- (a) Use the fact that probabilities sum to 1 to find  $q$ .
- (b) Compute  $P(X \geq 1)$  and  $P(X \neq 1)$ .
- (c) Could  $q = -0.10$  define a valid probability distribution? State the axiom that answers the question.

**10. Independent trials.** A fair coin is tossed five times.

- (a) Compare  $P(HHHHH)$  and  $P(HHTHH)$ . Compute both probabilities.
- (b) There are five different sequences with exactly one tail. Use this fact to compute the probability of exactly one tail.
- (c) Why does multiplying probabilities make sense here? What assumption are you using?

**Joint, Conditional, and Marginal Probability**

A joint distribution records probabilities for combinations of events. The table describes whether it rains ( $R$ ) and whether a person carries an umbrella ( $U$ ) on a randomly selected day.

|                | $R$  | $R^c$ | Marginal total |
|----------------|------|-------|----------------|
| $U$            | 0.30 | 0.10  |                |
| $U^c$          | 0.05 | 0.55  |                |
| Marginal total |      |       |                |

Recall that

$$P(B | A) = \frac{P(A \cap B)}{P(A)}, \quad P(A) = \sum_b P(A, b).$$

**11. Joint and marginal probabilities.**

- (a) Verify that the four joint probabilities form a valid distribution.
- (b) Fill in the row and column marginal totals in the table.
- (c) Use marginalization to compute  $P(U)$  and  $P(R)$ . Write each as a sum of joint probabilities.
- (d) Compute  $P(U \cap R)$  and describe this event in words.

**12. Conditional probability and independence.**

- (a) Compute  $P(R | U)$ . Interpret the result in a sentence.
- (b) Compute  $P(U | R)$ . Why is it different from  $P(R | U)$ ?
- (c) If  $U$  and  $R$  were independent, then  $P(U \cap R) = P(U)P(R)$ . Are they independent according to this table? Show your calculation.